

Shibani Singh

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SUMMARY

Machine Learning and Backend Engineer with 9+ years of software engineering experience designing production AI platforms, distributed ML infrastructure, and adversarially robust machine learning systems. Experienced in building enterprise-scale LLM workflows, retrieval systems, cloud and on-premise ML pipelines, and automation platforms deployed across thousands of users. Background spans AI security research, distributed systems, large-scale data processing, and applied machine learning.

SKILLS

Languages: Python, C++, C, Scala, Java, PHP

ML & AI: PyTorch, TensorFlow, LLMs, RAG, Embeddings, Vector Search, Adversarial ML, Model Evaluation, Distributed Training

Backend & Cloud: Python, AWS, Docker, Spark, Kafka, Linux, PostgreSQL, SQL

Domains: Machine Learning Engineering, ML Security, Distributed Systems, Big Data Analytics

EXPERIENCE

Lead Machine Learning Engineer, AI Platform

Jul 2023 – Present

Hilton, Platform Architecture & Automation (Remote)

- Designed and deployed a multi-agent LLM workflow platform that automated Marketing Tech QA for Hilton, orchestrating specialized AI agents for information retrieval, audit and format verification across all campaigns reducing manual effort by 80%.
- Designed, developed and deployed a production ML platform automating duplicate invoice detection and audit workflows across 100K+ invoices per month, reducing manual finance review by over 40% while improving anomaly detection precision.
- Designed and deployed a semantic search engine supporting 8,700+ Hilton properties, cutting RFP lookup time from hours to seconds and eliminating large volumes of repetitive work.
- Owned production ML pipelines from model development through deployment, integrating AWS services for embedding generation, retrieval evaluation, and enterprise administration workflows for reliable production delivery.

AI Applied Research Scientist

Oct 2020 – May 2023

Intel Labs, Security & Privacy Research Group

- Researched and implemented defenses against adversarial machine learning attacks under DARPA GARD, developing robustness evaluation pipelines and model hardening techniques against gradient and query-based attacks.
- Developed an open-source CARLA-based synthetic data generation framework enabling reproducible robustness benchmarking and adversarial evaluation across 10+ research teams.
- Published research at ICML workshops, collaborated with cross-functional ML teams on model reliability and evaluation tooling.

Cloud Solutions Engineer

Jun 2017 – Oct 2020

Intel Corporation, Data Center & AI Group

- Architected distributed machine learning workloads across Spark, Hadoop, and cloud infrastructure, improving training and inference performance by 2–4× for enterprise customers.
- Conducted large-scale benchmarking across on-prem and AWS, influencing multi-million dollar infrastructure decisions.
- Built a Spark-based deep learning pipeline analyzing millions of Reddit/Twitter posts for crypto price prediction; presented at Spark AI Summit (2018).

Software Engineering Intern

May 2016 – Aug 2016

Intel Corporation

- Developed ML algorithms for the Trusted Analytics Platform, improving distributed analytics throughput and model performance.

Software Engineering Intern

Jan 2014 – May 2014

Indian Institute of Technology (IIT) Bombay, MOOC Lab

- Built backend systems for an open-source e-learning platform and contributed to automated grading pipelines; presented work at LWMOOCs Workshop, MIT.

EDUCATION

M.S. in Cybersecurity

Aug 2023 – Dec 2025

New York University

Recipient of NYU Cyber Fellows Scholarship

GPA: 3.8 / 4.0

M.S. in Computer Science

2015 – 2017

Arizona State University

Eta Kappa Nu (IEEE Honors Society)

GPA: 3.8 / 4.0

Thesis: Deep learning based classification of FDG-PET data for Alzheimer's disease

B.Tech in Computer Science and Engineering

2011 – 2015

LNMICT Institute of Information Technology

GPA: 8.8 / 10.0

Thesis: Side-channel analysis of ECDSA in Bitcoin mining

PUBLICATIONS & AWARDS

Publications:

- ICML AdvML Frontiers: Synthetic Data for Adversarial ML (2022)
- Splunk .conf: High-Performance Analytics on Intel Architecture (2019)
- Spark AI Summit: Deep Learning for Bitcoin Price Prediction (2018)
- SPIE: Deep Learning-Based Classification of FDG-PET Data for Alzheimer's Disease (2017)
- Master's Thesis: Deep Learning-Based Classification of FDG-PET Data for Alzheimer's Disease Categories

Awards:

- 68th / 500 in Space Heroes CTF (2024)
- Intel Division Recognition Awards (2020)
- Full Funding Scholarship – MIT LWMOOCs Workshop (2014)